



SecureDial

Call Center Application

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Introduction

What is SecureDial?

A security-focused application for call centers to help protect sensitive client payment card information.

Uses Interactive Voice Response (IVR) system to interpret Dial-Tone Multi-Frequency (DTMF) inputs from a caller to verify payment card information.

What?

Interactive Voice Response (IVR) System

- the automated phone system that callers interact with.
- Ex. "Press 1 for more options"

Dial-Tone Multi-Frequency (DTMF)

- the tone system used when pressing keys on a phone keypad.

SecureDial uses this technology to receive payment card information from callers

- Independent
- Sensitive information never seen by agent

Why?

Many call centers still use live agents to operate and customers still strongly value access to live agents.

Many call centers still require people to read their sensitive card data aloud to verify

2023 CardEasy Market Research Report found

- 70% of customers were concerned about the security of their card details when communicated over the phone
- 70% of customers believe that using technology that reliably hides payment card details from contact center agents is secure

Company often liable when agents commit fraud with stolen payment card information

Background & Related Work

Background & Related Work

- Previously worked IT at a Call Center
- Very clear issue with how sensitive card data was being communicated
- My final essay for ENGL 415 (Written Communication for Engineers) was a recommendation report for the adoption of a DTMF system to protect client information at call centers

Project Requirements

Initial Project Requirements

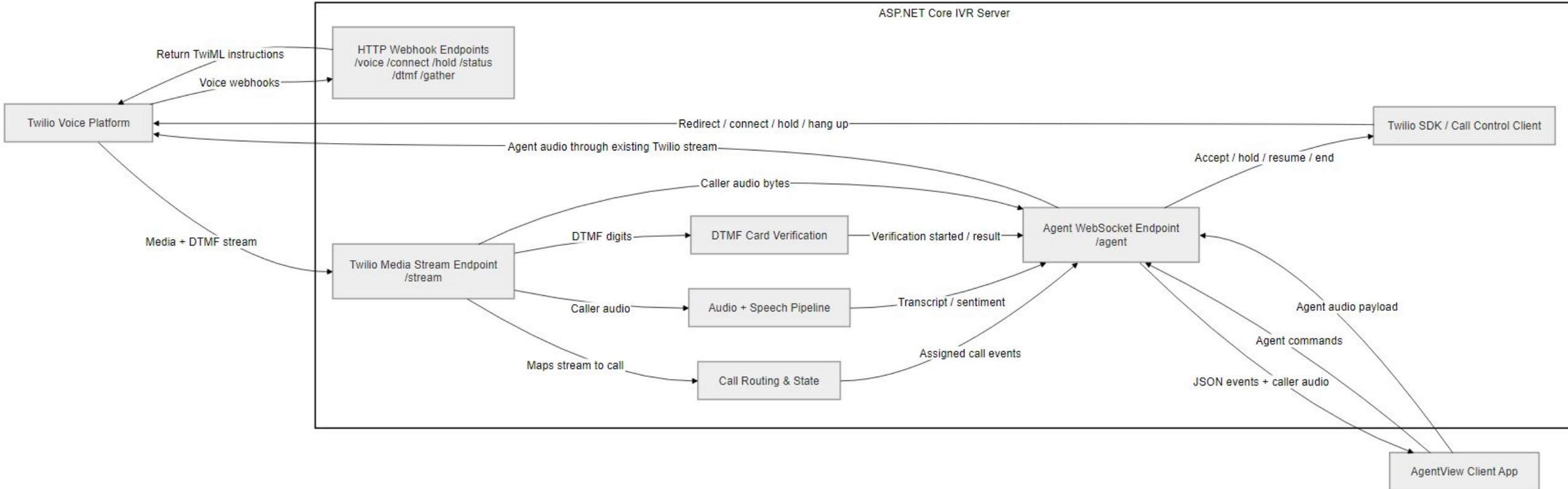
- MVP
 - Receive and interact with incoming phone calls
 - support live two-way audio between the phone caller and the agent
 - Interpret dial-tone frequencies as number inputs
 - Verify those number inputs as a valid payment card number
 - Hide payment card input information from the agent on the call
- Version 1.0:
 - Agent controls: Answering, hanging up, putting on hold.
 - Agent status: Available, On Call, Unavailable, etc.
 - Multiple agents available at once
- Version 2.0:
 - Call History
 - Manager View of agent status

Later Additions

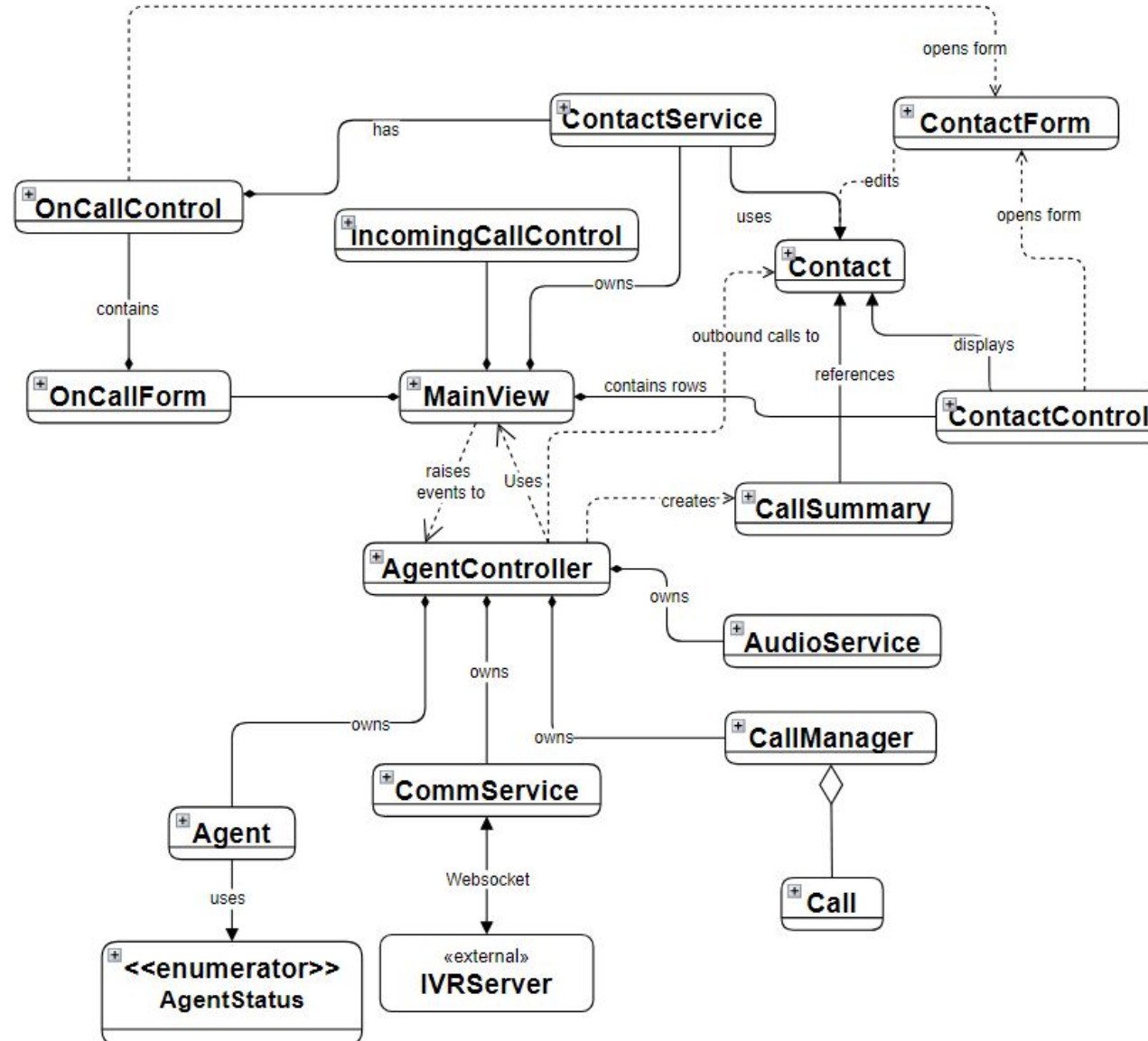
- Call Transcription
- Sentiment Analysis
- Contact Management

Project Design

Server Data Flow

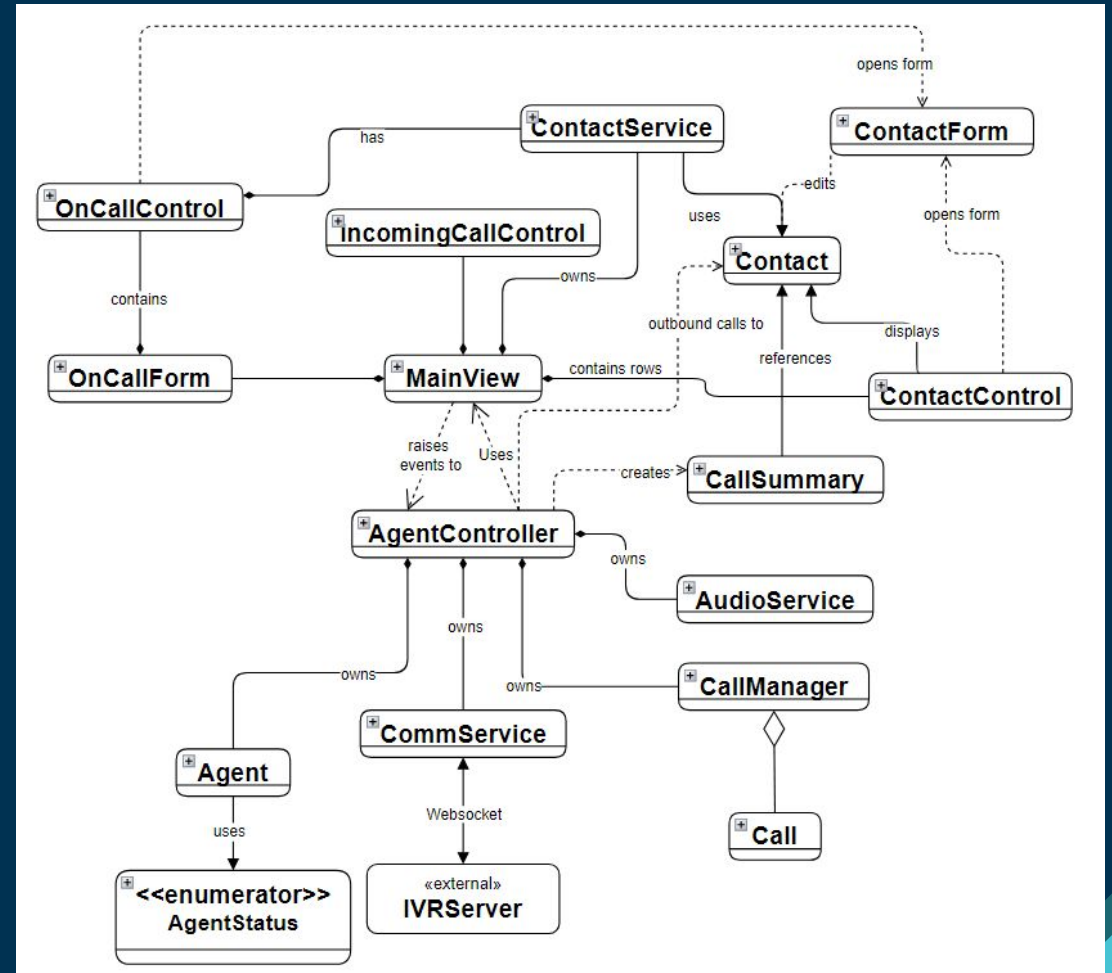


Frontend Class Diagram

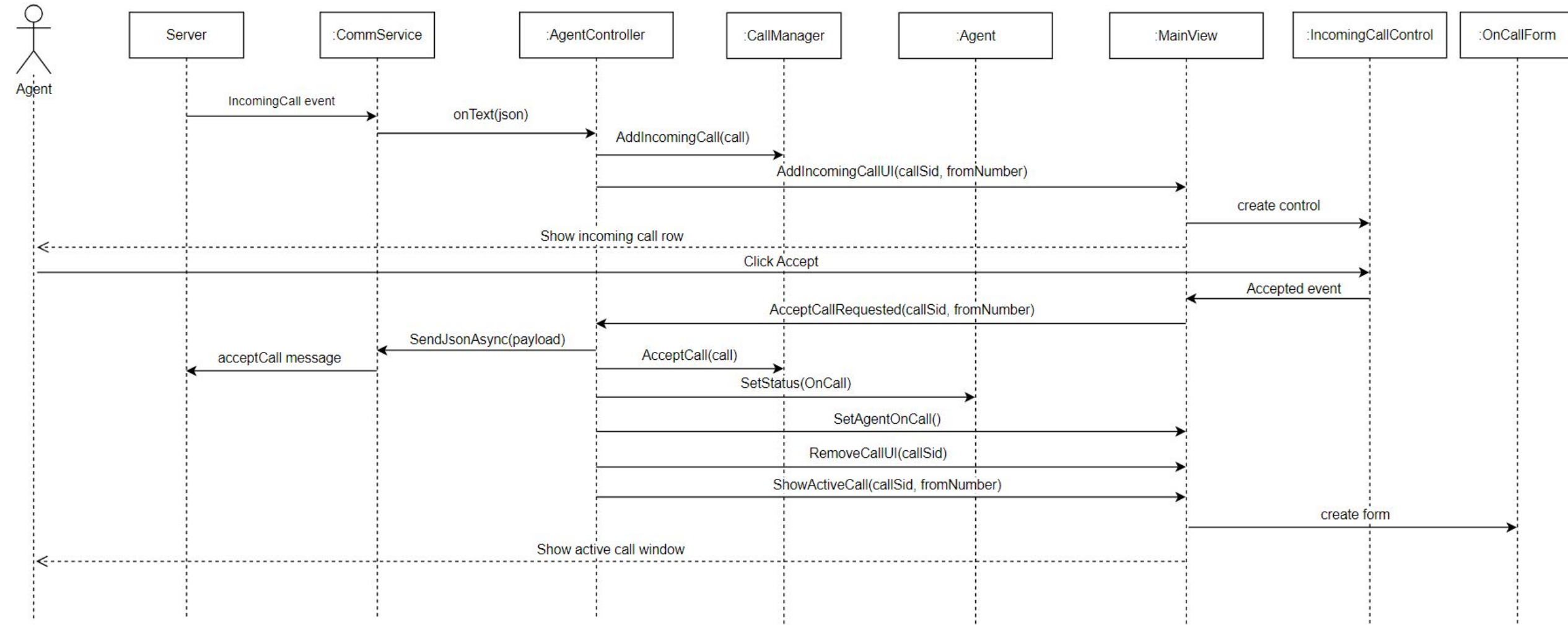


Code Structure / Separation of Responsibilities

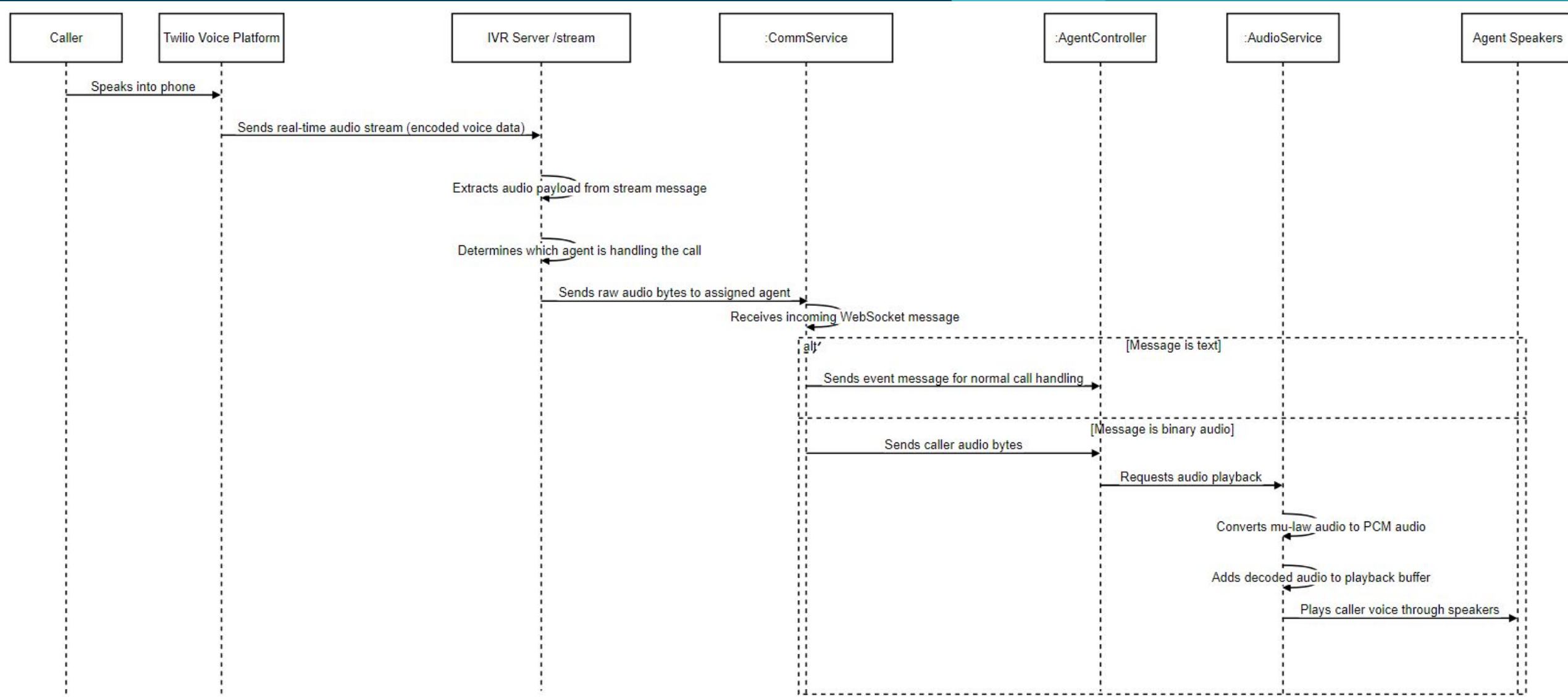
- MainView: UI display and menu switching
- AgentController: coordinates app logic.
- CommService: WebSocket communication
- AudioService: audio capture, playback, and conversion
- CallManager: tracks pending and active calls
- ContactService: contact persistence
- OnCallControl: active call buttons and controls



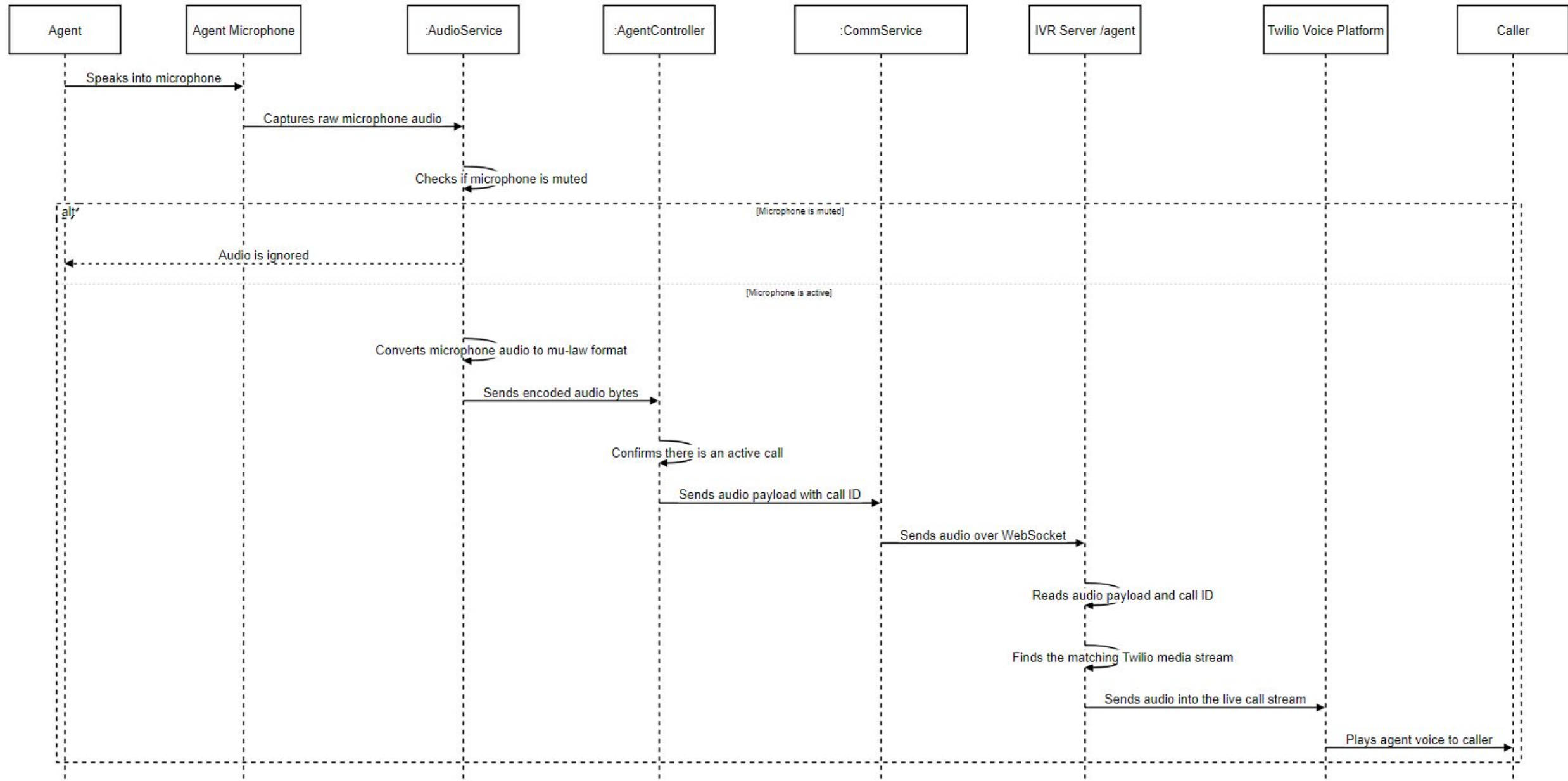
Sequence Diagram: Agent Accepts Phone Call



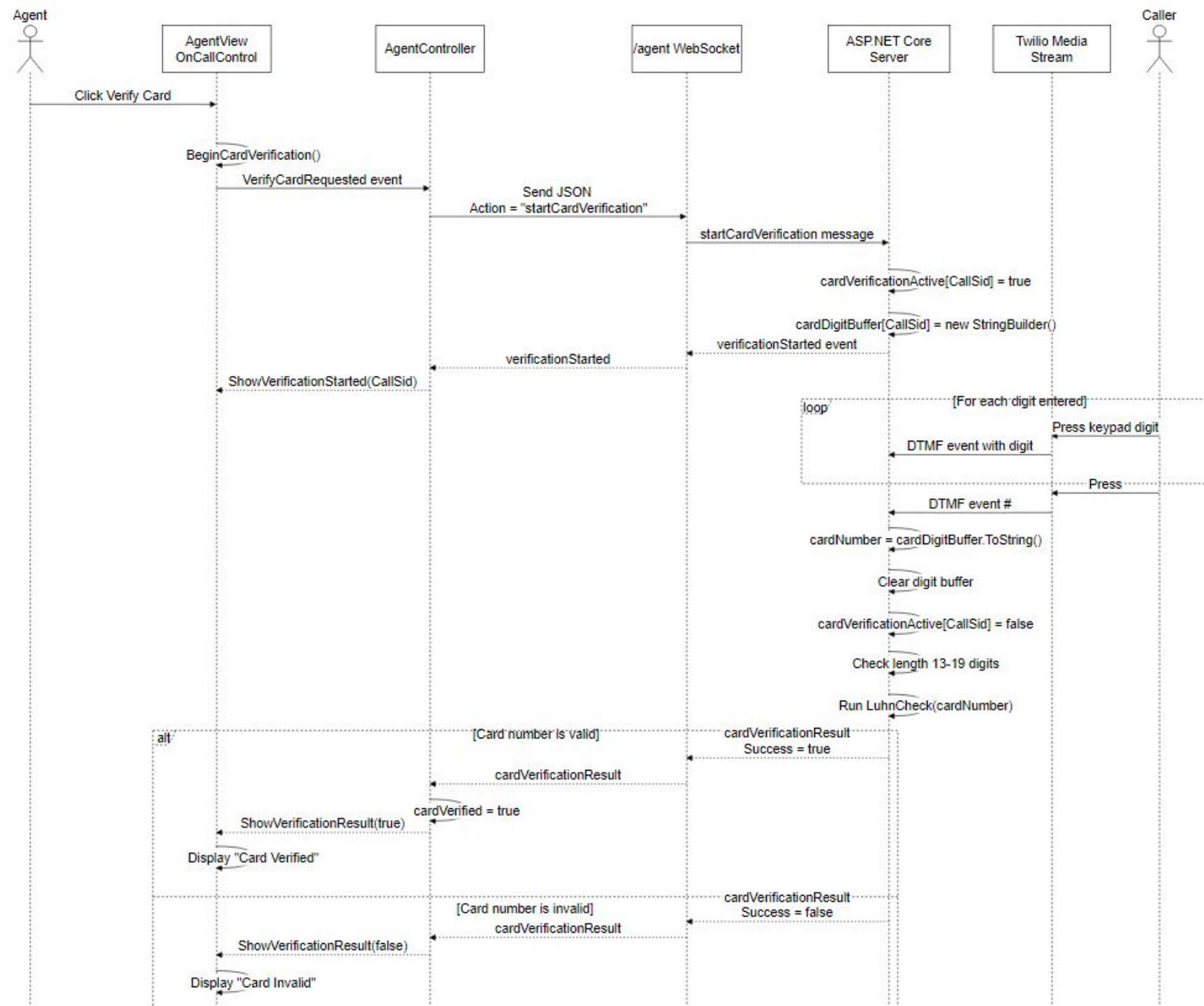
Sequence Diagram: Caller Audio Output in Agent Application



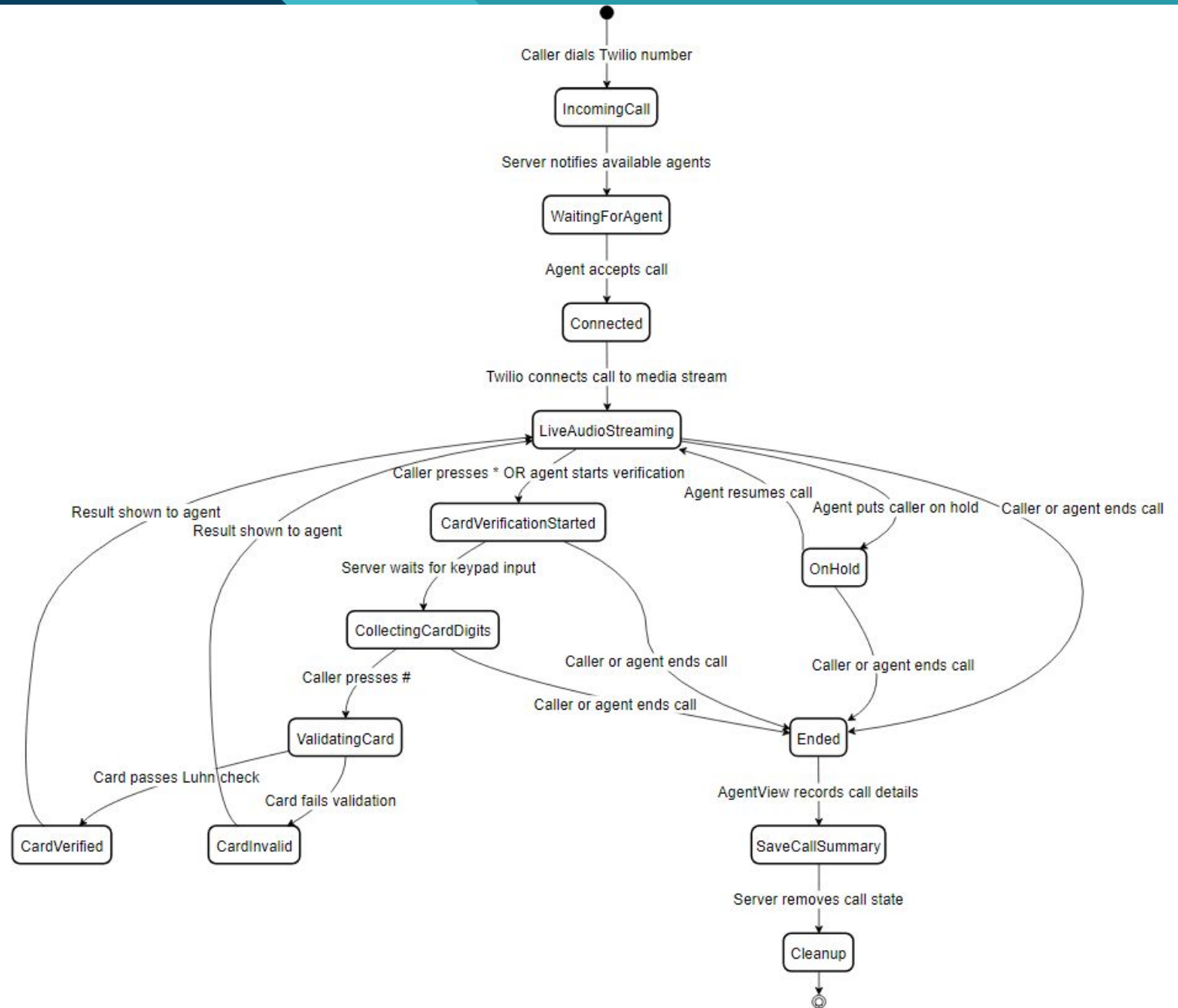
Sequence Diagram: Agent Audio Output in Caller Phone



Sequence Diagram: Agent Requests To Verify Card



Phone Call State Diagram



Project Implementation

Tech Stack



Backend:

C# ASP.NET
Core



Telephony:

Twilio, using
voice webhooks,
Twiml for call
controls



Audio
Processing:

NAudio, for mic
capture/speake
r playback



Speech
Recognition:

Vosk



Sentiment
Analysis:

VaderSharp2



Frontend UI:

Windows Forms

Websockets to communicate between server and app

Twilio

- Provides the real phone number
- Receives inbound calls
- Creates outbound calls
- Streams call audio
- Uses TwiML (Twilio Markup Language) to control the call state from the application



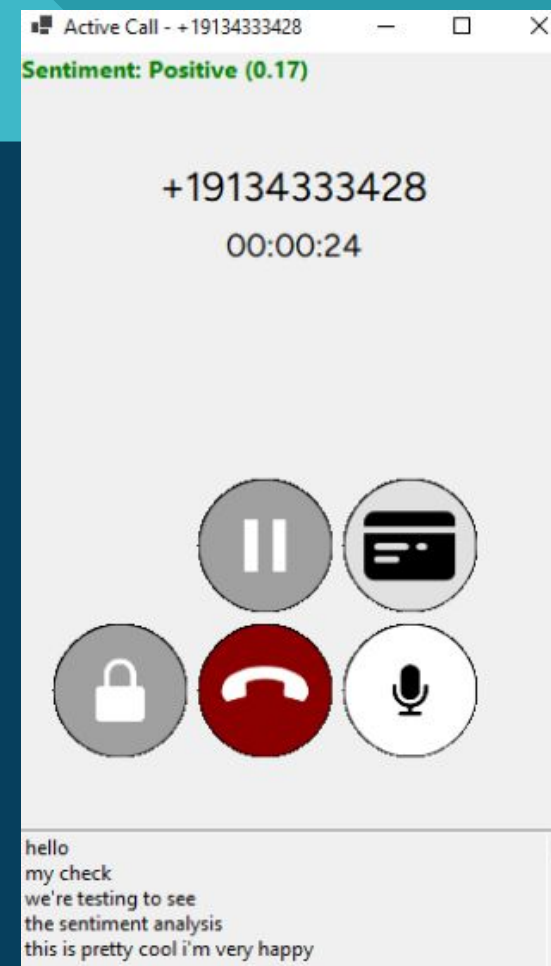
NAudio

- Handles microphone capture
- Speaker playback
- Buffering
- μ -law/PCM conversion
 - Twilio uses μ -law 8 kHz audio
 - Playback/transcription requires different audio format



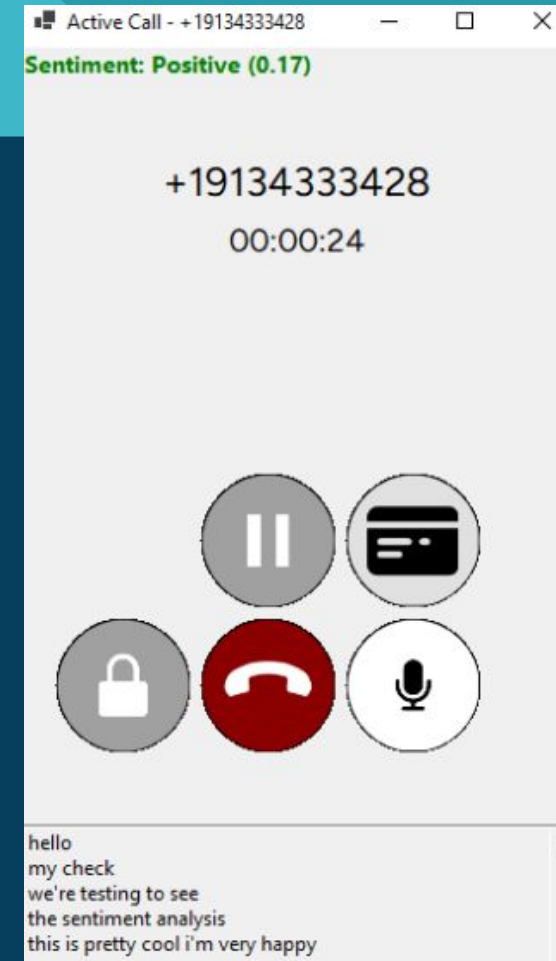
Vosk

- Performs speech-to-text transcription from the caller's audio
- runs locally/offline, expensive RAM usage
- Vosk produces partial and final transcription results
- Final transcript text is sent to the Application through a transcriptUpdate WebSocket message



VaderSharp2

- the sentiment analysis library used to evaluate the emotional tone of call transcripts
- a C#/.NET implementation of VADER, a rule-based sentiment analysis tool
- returns sentiment scores such as positive, neutral, negative, and a combined compound score



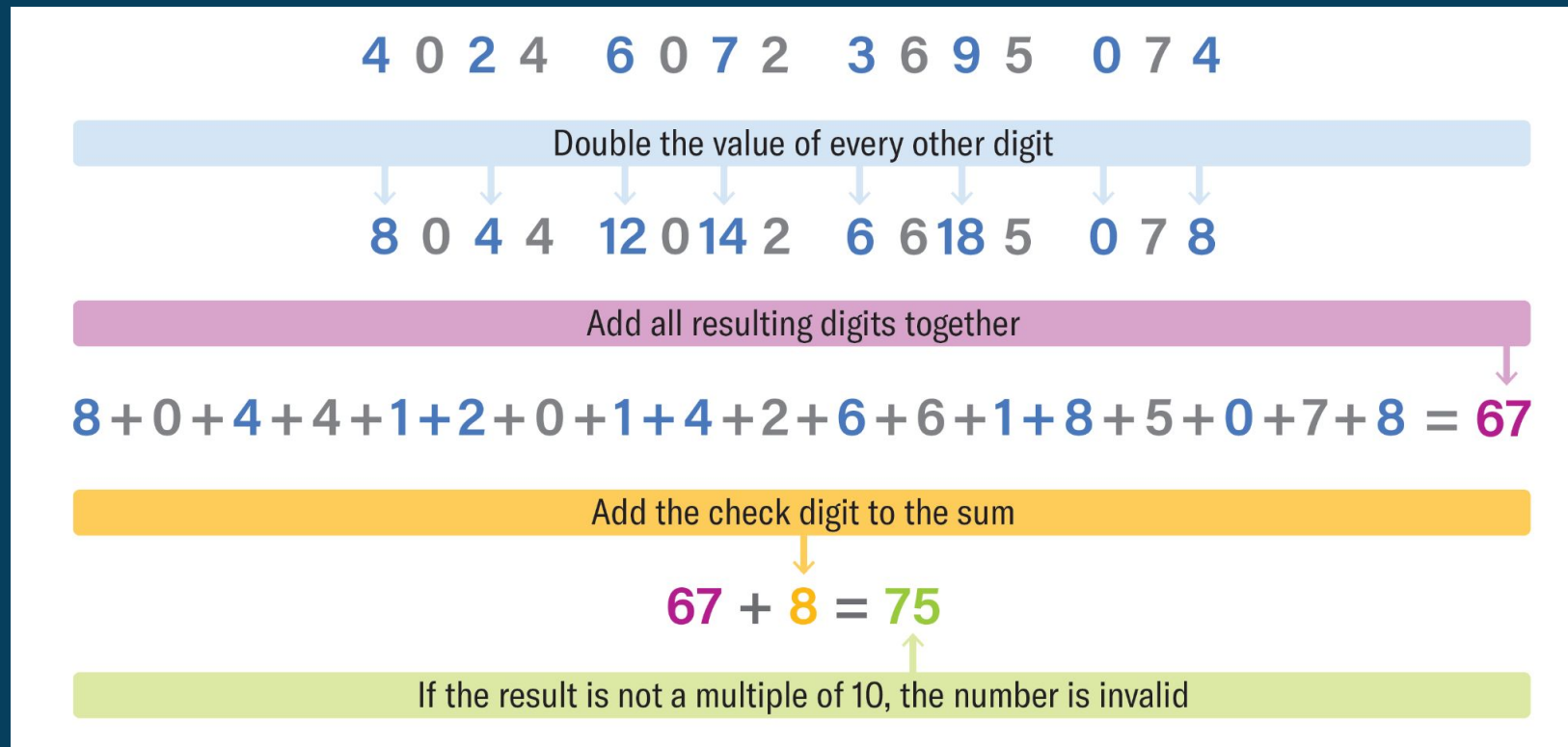
Ngrok

ngrok

- Ngrok exposes the locally running ASP.NET server to the public internet
- creates a public HTTPS URL that forwards traffic to the local server
- necessary because Twilio cannot send webhooks or media streams to localhost on the developer computer
- For development the server runs locally

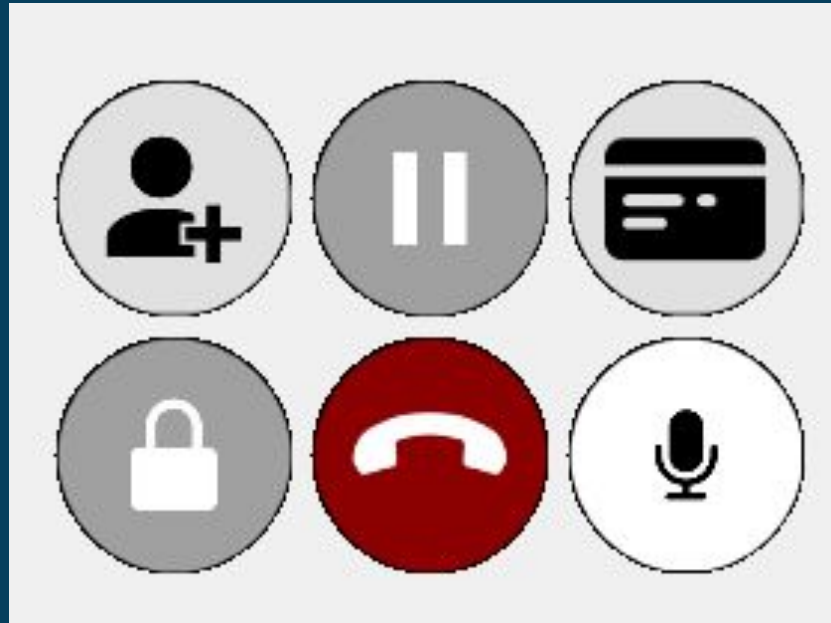
Card Verification

- Need way to verify card numbers without communicating with payment processors
- Luhn algorithm used to test if card number is structurally valid

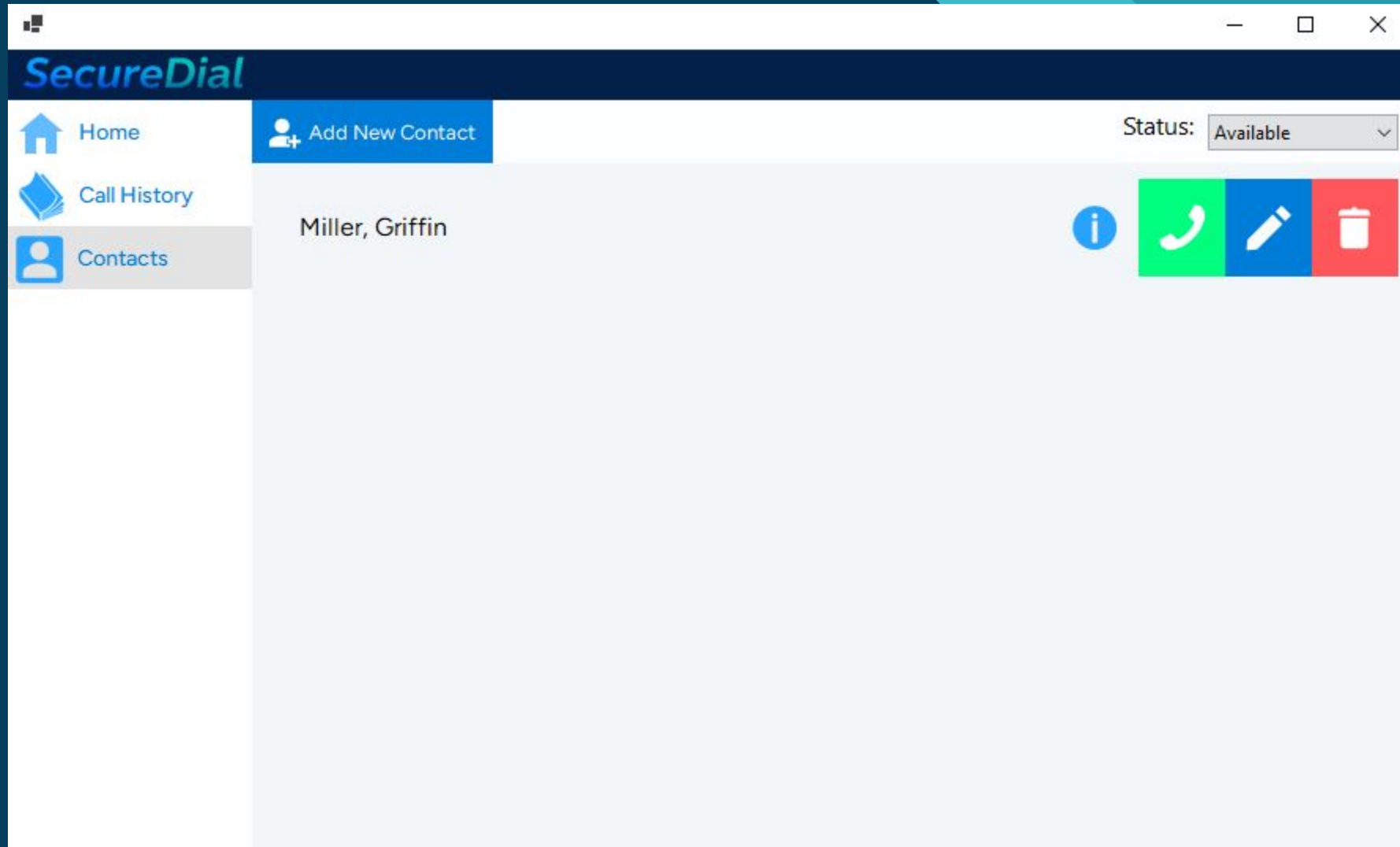


Call Control

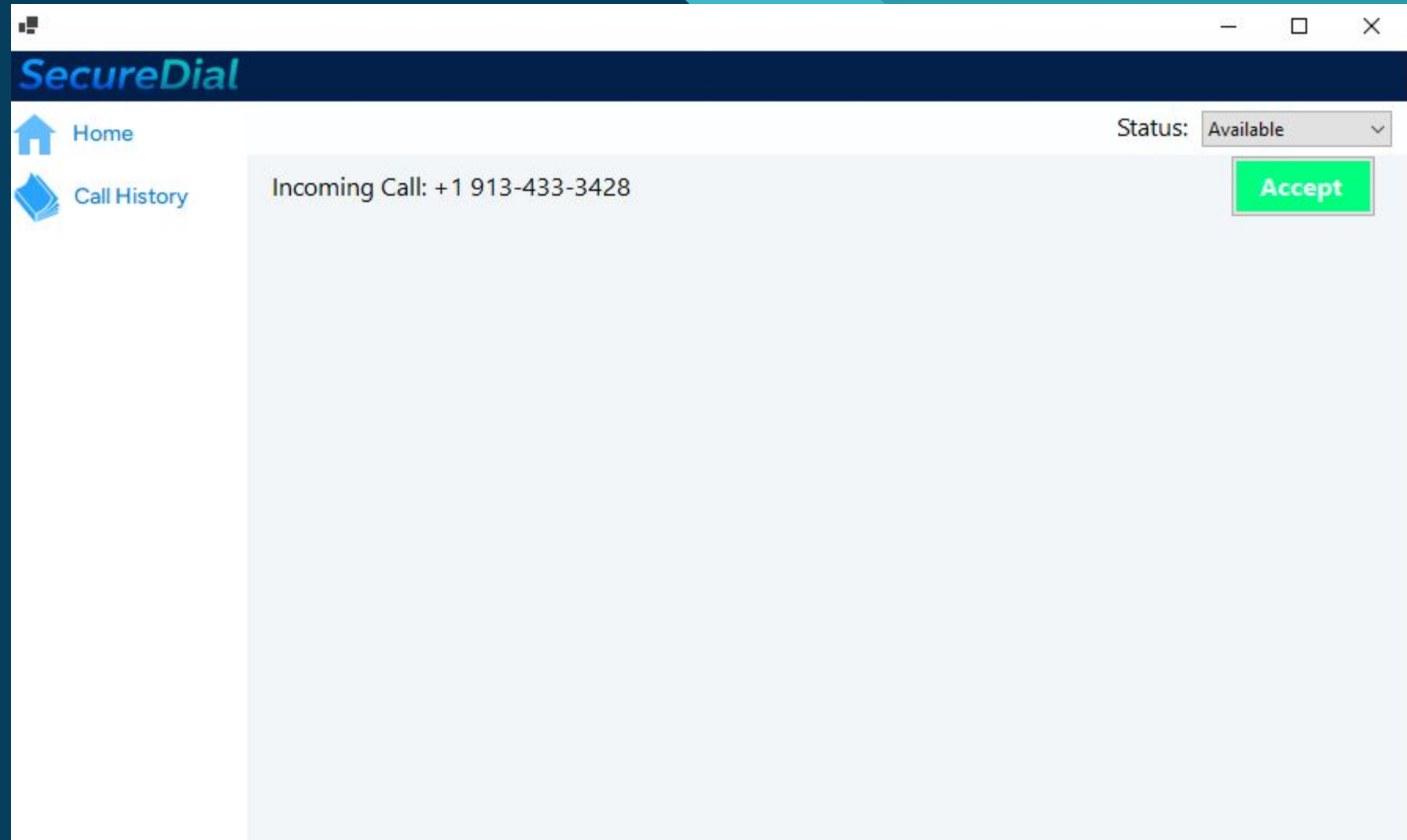
- Mute
- Hold
- Secure DTMF
- Begin DTMF listening
- Create Contact
- End Call



Contact Management



Incoming Call Control



Testing

Testing

- Tested edge cases often by calling from my own phone
- Had friends test the application and functionality

Deployment

Packaging & Deployment

- User needs
 - Visual Studio with .NET desktop development and ASP.NET workloads
 - Twilio Subscription
 - Twilio phone number
 - Public server URL or ngrok URL, replace URLs in server and one string in application code for the connection URL
 - Set these environment variables:
 - TWILIO_ACCOUNT_SID
 - TWILIO_AUTH_TOKEN
 - TWILIO_PHONE_NUMBER
- Twilio webhooks must point to the deployed server
- Run server, then run app
- Current system is functional for demo, impractical for a release build



Demo

Future Work

Future Work

- Replace json files for contacts and call logs with a database structure
- ngrok should be replaced with hosted server and permanent domain name
- Configuration system
 - Remove hardcoded URLs, Twilio numbers, and ngrok addresses
- Better transcription tool
 - Current tool requires significant RAM usage for better model

Conclusion

Functionality Review

- Receiving incoming calls
- Live two-way audio
- Call interaction
- DTMF Card Verification
- Call History logs
- Transcription
- Sentiment analysis
- Contact management
- Outbound calling



Questions?